

# Hyojoon Park

PH.D. CANDIDATE | 3D GRAPHICS/VISION, ML/COMPRESSION/SIMULATION

hpark376@wisc.edu | [Personal Website](#) | [GitHub](#) | [LinkedIn](#) | [Google Scholar](#)

## Professional Summary

- My research integrates **Machine Learning** with **Computer Graphics** and **Computer Vision**, building end-to-end systems that combine physics-based simulation, geometry, and image/video processing.
- My work spans **3D reconstruction** of bodies, faces, and scenes, from camera calibration and bundle adjustment to **3D Gaussian Splatting (3DGS)**, along with **neural simulation** of face and hair, hand reconstruction for **XR (VR/AR)**, and **haptics** for dental simulation.
- I am currently developing **4D medical image and video compression with deep learning** using spatiotemporal wavelets, vision transformers (ViT), hyperprior entropy models, latent quantization, and rate-distortion optimization.

## Education

### University of Wisconsin-Madison

Madison, WI, USA

Ph.D. Candidate in Computer Sciences

Sep. 2021 - Expected Dec. 2026

- **Research:** Integration of ML with physics-based simulation for computer graphics and neural compression for medical imaging
- **Advisors:** Eftychios Sifakis and Kevin Elceiri

### Seoul National University

Seoul, Korea

M.S. in Mechanical Engineering

Mar. 2017 - Feb. 2019

- **Thesis:** Dental Simulator with Increased Z-width of Haptic Rendering (also presented at AsiaHaptics 2018)
- **Advisor:** Dongjun Lee
- **Outstanding MS Thesis Presentation Award** [[M.S. thesis presentation](#)]

#### Technical Contributions

- Extended **passive decomposition** to SE(3) using **passive midpoint integration** for stable, stiff haptic rendering.
- Derived a **momentum-based human force observer** to estimate human forces on multi-DoF haptic devices without force sensors.
- Implemented **octree voxel rendering** for real-time collision detection and material removal in virtual tooth drilling.

### Korea University

Seoul, Korea

B.S. in Mechanical Engineering

Mar. 2010 - Feb. 2017

- Apr. 2014 - Jul. 2014: **Technical University of Munich (TUM)** - Exchange Student
- Jun. 2011 - Mar. 2013: **Republic of Korea Army (ROKA)** - Military Service

## Work Experience

### Google

Playa Vista, CA, USA

Student Researcher

Sep. 2025 - May 2026

- Developed **real-time hair animation** for **digital humans** and **XR** with a **self-supervised, physics-based neural hair simulator**. Publication-track results in progress.

#### Technical Contributions

- Designed a **recurrent (GRU) motion encoder** to model hair inertia, natural swinging, and settling from motion history.
- Encoded local and global hair-groom structure with a **latent transformer** that scales linearly with strand count.
- Built a **StyleGAN2**-inspired **modulated MLP** decoder conditioned on body shape, hairstyle, and motion history for hair animation.

### Nokia Bell Labs

New Providence, NJ, USA

Industrial Metaverse Intern

Jun. 2025 - Aug. 2025

- Built a real-time 3DGS **digital twin** with **language-guided 3D interaction** and physical registration, placing **first among 100+ global interns**. Follow-up research studies capture-geometry effects on reconstruction quality. Publication-track results in progress.

#### Technical Contributions

- Registered 3DGS to real-world coordinates using ChArUco triangulation, reprojection optimization, and Umeyama alignment.
- Integrated **CLIP** features and **SAM** masks into 3D Gaussians with DBSCAN for real-time **text-guided 3D segmentation**.

### NVIDIA

Santa Clara, CA, USA

Software Intern - Physics-Based Simulations

May 2024 - Aug. 2024

- Developed **single-image 3D face reconstruction** for **identity-preserving digital humans**, re-aging faces from infancy to old age.

#### Technical Contributions

- Generated age-specific training meshes through **deformation transfer** from scanned faces to age-specific templates.
- Learned the **3D morphable model (3DMM)** identity basis from skull-aligned face meshes centered on age-specific reference shapes.
- Trained an end-to-end network to predict 3DMM identity and age parameters from a single image using a pretrained MiVOLO age prior, **differentiable rendering**, and **GAN-based training**.

## University of Utah

### Graduate Research Assistant

Salt Lake City, UT, USA

Sep. 2019 - Jun. 2021

- Developed and open-sourced a multi-view body-reconstruction system for SIGGRAPH 2021, recovering 1,000+ uniquely labeled points from RGB cameras and a low-cost patterned suit to capture fine skin deformation. [\[code\]](#) [\[paper\]](#)

#### Technical Contributions

- Built a **multi-camera calibration and bundle adjustment** pipeline in C++ and Python with Ceres Solver.
- Built a **VAE outlier detector** to filter erroneous checkerboard corners and improve calibration.
- Trained **CNN regressors and classifiers** for corner localization, candidate validation, and code recognition.

## Korea Institute of Science and Technology (KIST)

### Research Intern

Seoul, Korea

Mar. 2019 - Aug. 2019

- Developed an IROS 2020 VR hand-rendering system that reconstructs realistic full-hand poses from fingertip positions. [\[paper\]](#) [\[video\]](#)

#### Technical Contributions

- Calibrated virtual hand size from fingertip measurements and tracking-device geometry using SVD-based similarity alignment.
- Applied **motion retargeting** with per-joint scaling and offsets calibrated from open-hand and grasping poses.
- Extended **FABRIK inverse kinematics** with planar hinge constraints for natural finger bending during **real-time hand reconstruction**.

## Publications

---

### Near-realtime Facial Animation by Deep 3D Simulation Super-Resolution

**Hyojoon Park**, Sangeetha Grama Srinivasan, Matthew Cong, Doyub Kim, Byungsoo Kim, Jonathan Swartz, Ken Museth, Eftychios Sifakis  
ACM Transactions on Graphics (TOG), 2024 (Presented at SIGGRAPH ASIA 2024) [\[paper\]](#) [\[code\]](#) [\[featured video\]](#)

- Combines coarse simulation with **neural 3D super-resolution**, deployed with **ONNX** to produce detailed facial animation at 18.46 FPS (115x faster than the traditional simulator).
- Uses **graph neural networks** and **coordinate-based MLPs** for volumetric encoding and geodesic interpolation of coarse simulation output, generalizing from quasi-static training data to unseen expressions, dynamics, and external forces.

### Collagen Fiber Centerline Tracking in Fibrotic Tissue via Deep Neural Networks with Variational Autoencoder-based Synthetic Training Data Generation

**Hyojoon Park\***, Bin Li\*, Yuming Liu, Michael S. Nelson, Helen M. Wilson, Eftychios Sifakis, and Kevin W. Eliceiri (\*equal contributions)  
Medical Image Analysis, 2023 [\[paper\]](#) [\[code\]](#)

- Introduces **DuoVAE**, pairing two independent VAEs for **controllable synthetic data generation** across diverse collagen-fiber morphologies.
- Combines a **conditional GAN** for image synthesis with **U-Net** centerline extraction for collagen-fiber centerline tracking.

### Capturing Detailed Deformations of Moving Human Bodies

He Chen, **Hyojoon Park**, Kutay Macit, and Ladislav Kavan

SIGGRAPH, 2021 [\[paper\]](#) [\[code\]](#)

- Reconstructs 1,000+ uniquely labeled body points from 16 RGB cameras using **CNN regressors and classifiers** for corner localization, candidate validation, and code recognition, followed by **multi-view triangulation**, without temporal tracking for point identification.
- Captures breathing, muscle motion, and flesh deformation with pixel-level reprojection accuracy through camera calibration and **bundle adjustment**.

### Adaptive Precision-Enhancing Hand Rendering for Wearable Fingertip Tracking Devices

**Hyojoon Park** and Jung-Min Park

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020 [\[paper\]](#) [\[video\]](#)

- Extends **FABRIK inverse kinematics** with planar hinge constraints for real-time full-hand reconstruction from fingertip positions and uses **motion retargeting** to improve pose alignment.
- Reduces grasping-task time by 17.08% for large objects and 28.14% for small objects versus rendering the hand as spheres.

### Dental Simulator with Increased Z-width of Haptic Rendering

**Hyojoon Park**, Myungsin Kim, and Dongjun Lee

AsiaHaptics, 2018 [\[paper\]](#) [\[video\]](#)

- Uses **passive midpoint integration** to substantially increase stable virtual-coupling stiffness for high-fidelity haptic rendering.
- Extends **passive decomposition** to 6-DoF haptics with a momentum-based human force observer, reducing device-proxy coordination error from 20 mm to near zero.

## Publications (continued)

---

### **Rigid-body Collaborative Manipulation among Remote Users with Wearable Cutaneous Haptic Interfaces**

Myungsin Kim, WonHa Lee, **Hyojoon Park**, Junghan Kwon, Yong-Lae Park, and Dongjun Lee

AsiaHaptics, 2018 [[paper](#)] [[video](#)]

- Enables two remote users to manipulate a shared rigid body in VR, estimating finger motion from **soft sensors** and **IMUs** with three-DoF cutaneous fingertip feedback.

### **Stretchable Skin-Like Cooling/Heating Device for Reconstruction of Artificial Thermal Sensation in Virtual Reality**

Jinwoo Lee, Heayoun Sul, Wonha Lee, Kyung Rok Pyun, Inho Ha, Dongkwan Kim, **Hyojoon Park**, Hyeonjin Eom, Yeosang Yoon, Jinwook Jung, Dongjun Lee, and Seung Hwan Ko

Advanced Functional Materials, 2020 [[paper](#)]

- Reproduces hot, cold, and material-dependent thermal cues at three fingertips in VR using stretchable heating/cooling devices.

### **Highly Stretchable and Oxidation-resistive Cu Nanowire Heater for Replication of the Feeling of Heat in a Virtual World**

Dongkwan Kim, Junhyuk Bang, Wonha Lee, Inho Ha, Jinwoo Lee, Hyeonjin Eom, Myungsin Kim, Jungjae Park, Joonhwa Choi, Jinhyung Kwon, Seungyong Han, **Hyojoon Park**, Dongjun Lee, and Seung Hwan Ko

Journal of Materials Chemistry A, 2020 [[paper](#)]

- Reproduces contact and radiative heat in VR using a 12-pixel thermal glove with motion tracking and closed-loop temperature control.

### **Wearable Cutaneous Haptic Interface with Soft Sensors and IMUs**

Yongjun Lee, Myungsin Kim, Yongseok Lee, **Hyojoon Park**, and Dongjun Lee

Korea Robotics Society Annual Conference, 2018

### **Design and Performance Evaluation of Wearable Haptic Interfaces**

WonHa Lee, Myungsin Kim, **Hyojoon Park**, and Dongjun Lee

International Conference on Control, Automation and Systems, 2018

## Professional Service

---

SIGGRAPH Asia 2026 Reviewer

2026

## Technical Skills

---

|                   |                                                                                                                                                                                                                                               |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages</b>  | Python, C++, C, CUDA                                                                                                                                                                                                                          |
| <b>Frameworks</b> | PyTorch, JAX, ONNX, OpenCV, NumPy/SciPy, OpenGL, Ceres Solver                                                                                                                                                                                 |
| <b>Core Areas</b> | Machine Learning, Deep Learning, Computer Vision, Computer Graphics, 3D Vision, 3D Reconstruction, Point Cloud Processing, 3D Segmentation, Neural Rendering, Physics-based Simulation, Medical Imaging, Image/Video Compression, XR, Haptics |
| <b>Methods</b>    | Transformers, Vision Transformers (ViT), GNNs, CNNs, RNNs (GRU), MLPs, GANs, VAEs, StyleGAN2, U-Net, CLIP, SAM, 3DGS, Differentiable Rendering                                                                                                |
| <b>Tooling</b>    | Linux, Git, Docker, CMake, LaTeX, Unity, Blender                                                                                                                                                                                              |
| <b>Hardware</b>   | HTC VIVE (headset, tracker, base station), 3D Systems Phantom Premium haptic device, multi-camera capture rigs                                                                                                                                |

## Teaching Experience

---

|                                                                            |             |
|----------------------------------------------------------------------------|-------------|
| Computer Graphics (CS559) <i>University of Wisconsin-Madison</i>           | Spring 2022 |
| Computer Graphics (CS559) <i>University of Wisconsin-Madison</i>           | Fall 2021   |
| Interactive Computer Graphics (CS6610) <i>University of Utah</i>           | Spring 2021 |
| Computer Graphics (CS4600) <i>University of Utah</i>                       | Fall 2020   |
| System Analysis in Mechanical Engineering <i>Seoul National University</i> | Spring 2018 |